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Cybersecurity: The Next 20 Years



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As widely discussed, acknowledged and debated on various forums across the world, we are living in a digital age, driven by technologies that didn't exist two decades ago. While a few technologies did exist in some form, they have matured over the years, and converged with one another and with newer more advanced technologies.

A simple example is the mobile phone. In the late 90s they were mere calling and text messaging devices, but still a defining technology then. The cellphone's journey to becoming a smartphone is essentially about integration of more and more layers of technology on the device, from cameras, improved computing, to 4G connectivity, artificial intelligence capabilities and a lot more.

In 2041, we can only imagine what looking back at today will reveal. Like the novelty of cellphones about 20 years ago, every decade has a defining novel technology which will evolve and catalyse a new innovation for the next decade. While smartphones are taken for granted today, we have novel tech-

nologies like Artificial Intelligence that have just embarked on their journey to maturity. In the area of networks and connectivity, 5G has become a reality, and will fuel a revolution in Internet of Things (IoT) use cases. The amalgamation of various technologies is resulting in a data explosion, and as organisations look at extracting meaningful insights from this data, AI and Machine Learning tools come to their rescue.

We are witnessing how these technologies have paved their way into our lives: Smart cities are getting smarter with increasing IoT applications; cloud based communication has received an immense fillip, especially in a remote centric world; cloud computing is happening at scale; drones are raising the bar of innovation and opening doors to new possibilities; virtual and augmented reality are making inroads into education, skill development, gaming and e-commerce, whereas breakthroughs in autonomous transportation are aiming at transforming the future of mobility. Given the pace of innovation and evolution of real life use cases, the next few years will see significant disruption.

According to a Gartner prediction, more than one million drones will be making deliveries in 2026, and CIOs in the mobility and transportation industries will be forced to re-evaluate their business and operational models. By 2041, drones could be a part of a larger hyper-connected logistical network, operating on global geographies. This would open a new arena of target for cyber attackers with the intent of jeopardising movement of critical goods, theft of critical materials, smuggling of prohibited goods or piracy.

As autonomous trucks hit the roads carrying goods like their aerial counterparts and as people commute in driverless cabs, criminals will try to infiltrate the mobility infrastructure – to bring an entire city's mobility network to a standstill, or to target individuals or steal goods.

Physical identity identification and authentication at ports and transit points could be replaced partly or completely by facial recognition technologies and biometrics. While this will greatly streamline the travel experience, criminal networks would seek ways to manipulate algorithms to compromise security and carry out a larger criminal activity.

AI will also significantly change the way customers consume services. Voice based conversational AI will transform customer interactions and drive the overall customer experience. As AI interference in customer interactions increase, a huge pool of a new set of data would be generated which will need to be protected as well.

5G will lead to massive growth of hyper-connected IoT devices, and that will fuel data generation at the Edge. In order to make smart cities smarter, the city's IT infrastructure will not just generate massive amounts of data, but all elements of this connected infrastructure will leverage insights from each other's datasets. In a digital environment like this, any vulnerability holds the potential to wreak havoc and bring an entire city to a standstill. Not just cities, but advances in digitalisation will lead to a dramatic rise in nation-state attacks. Great emphasis will be put by governments to protect their country's digital sovereignty.

While each of the above technologies will constantly evolve in their own field, they will also be weaved together intricately to function as a digital ecosystem. Cybersecurity practices will need to be holistic, because a breach in one component could potentially pose a risk to the entire IT infrastructure of organisations, cities or nations! We must build our cybersecurity infrastructure in a way that enables every layer of security to learn from threat intelligence generated by the other, adding protection at each level of development and aim towards achieving Cyber Immunity by 2041. 